

Solutions — Paper A

Solution 1 (10 marks). Apply Chapter 2's forward rate formula:

$$L(0; 4, 4.5) = \frac{1}{\tau} \left(\frac{P(0, 4)}{P(0, 4.5)} - 1 \right) = \frac{1}{0.5} \left(\frac{0.8530}{0.8365} - 1 \right)$$

$P(0, 4)/P(0, 4.5) = 1.019725$, so

$$L = \frac{0.019725}{0.5} = 0.039450 \rightarrow \boxed{3.945\%}$$

(5 marks: correct formula and ratio; 5 marks: correct final number to at least 3 significant figures.) A common error worth 0 method marks: using τ in the denominator of the ratio itself rather than only outside the bracket.

Solution 2 (10 marks). The annuity (Chapter 4):

$$A(0) = \sum_{i=1}^4 \tau_i P(0, T_i) = 0.9700 + 0.9350 + 0.8960 + 0.8540 = 3.6550$$

(6 marks: correct sum, shown explicitly.) The par rate, via the telescoping form:

$$S(0) = \frac{P(0, 0) - P(0, 4)}{A(0)} = \frac{1 - 0.8540}{3.6550} = \frac{0.1460}{3.6550} = 0.039945 \rightarrow \boxed{3.995\%}$$

(4 marks: correct formula and final number.) Full marks also awarded for computing $S(0)$ via the sum-of-forward-rates definition directly, provided the individual forward rates are shown and correctly derived from the same discount factors.

Solution 3 (12 marks). Bachelier's formula (Chapter 6):

$$d = \frac{F - K}{\sigma_n \sqrt{T_1}} = \frac{0.0395 - 0.0375}{0.0082 \times \sqrt{2.0}} = \frac{0.0020}{0.011597} = 0.17247$$

(3 marks: correct d .)

$$\Phi(0.1725) = 0.5685, \quad \varphi(0.1725) = 0.3931$$

(3 marks: both values correct to 3 d.p.)

$$d \Phi(d) + \varphi(d) = 0.17247 \times 0.5685 + 0.3931 = 0.09807 + 0.3931 = 0.4911$$

(2 marks.)

$$\text{Caplet} = N \tau P(0, T_2) \sigma_n \sqrt{T_1} \times 0.4911 = 25,000,000 \times 0.25 \times 0.9280 \times 0.011597 \times 0.4911$$

$$\approx \boxed{\$33,030}$$

(4 marks: correct final assembly and number, within rounding tolerance of $\pm \$50$.)

Solution 4 (8 marks). Cap–floor parity (Chapter 5):

$$\text{Caplet} - \text{Floorlet} = N\tau P(0, T_2) (F - K) = 25,000,000 \times 0.25 \times 0.9280 \times 0.0020 = \$11,600$$

(4 marks: correct parity difference.)

$$\text{Floorlet} = \text{Caplet} - \$11,600 = \$33,030 - \$11,600 = \boxed{\$21,430}$$

(4 marks: correct subtraction and final number, consistent with the candidate's own Question 3 answer.) No marks awarded for re-deriving the floorlet via a fresh Bachelier calculation — the question specifically credits recognising and using the parity shortcut.

Solution 5 (12 marks). ATM Bachelier shortcut (Chapter 7):

$$C_n(S) = \sigma_n \sqrt{T/2\pi} = 0.0088 \times \sqrt{2} \times \frac{1}{\sqrt{2\pi}}$$

Using $\sqrt{2/2\pi} = \sqrt{1/\pi} = 0.56419$:

$$C_n(S) = 0.0088 \times 0.56419 = 0.004965 \rightarrow 49.65 \text{ bp of rate}$$

(6 marks: correct rate-unit premium, shown explicitly as its own step.)

$$V = N \times A(0) \times C_n(S) = 30,000,000 \times 4.35 \times 0.004965 \approx \boxed{\$647,900}$$

(6 marks: correct final cash premium.) A common partial-credit error: forgetting the $\sqrt{T}$ inside $C_n(S)$ and using $\sigma_n/\sqrt{2\pi}$ directly — this earns method credit for recognising the ATM shortcut structure but not the final numerical marks.

Solution 6 (14 marks). Since $1 + \tau_j L_j(t) = P(t, T_j)/P(t, T_{j+1})$, apply Itô's lemma to $Y_j(t) := \log(1 + \tau_j L_j(t))$:

$$dY_j = \frac{\tau_j dL_j}{1 + \tau_j L_j} - \frac{1}{2} \frac{\tau_j^2 (dL_j)^2}{(1 + \tau_j L_j)^2}$$

(4 marks: correct application of Itô's lemma to the log.) Substituting $dL_j = \sigma_j L_j dW_j^{j+1}$, the diffusion (volatility) term of Y_j — the only part relevant to Girsanov's drift-correction formula, since drift terms don't contribute to the volatility of a numeraire ratio — is

$$\text{vol of } Y_j = \frac{\tau_j \sigma_j L_j}{1 + \tau_j L_j}$$

(6 marks: correctly isolating the diffusion coefficient, discarding the $(dL_j)^2$ drift-contributing term for the purpose of stating the volatility.) Hence

$$\gamma_j(t) = \frac{\tau_j \sigma_j(t) L_j(t)}{1 + \tau_j L_j(t)}$$

(4 marks: correctly stated final answer, matching Chapter 13's key result box exactly.)

Solution 7 (14 marks). First the $j = 1$ term (Chapter 13's terminal-measure formula):

$$\gamma_1 = \frac{\tau \sigma_1 L_1}{1 + \tau L_1} = \frac{0.5 \times 0.21 \times 0.0380}{1 + 0.5 \times 0.0380} = \frac{0.003990}{1.0190} = 0.003916$$

$$\text{term}(j=1) = -\sigma_0 L_0 \rho_{01} \gamma_1 = -0.19 \times 0.0360 \times 0.88 \times 0.003916$$

$$= -2.357 \times 10^{-5}$$

(6 marks: correct γ_1 and $j = 1$ term.) Then the $j = 2$ term:

$$\gamma_2 = \frac{0.5 \times 0.23 \times 0.0395}{1 + 0.5 \times 0.0395} = \frac{0.004543}{1.01975} = 0.004455$$

$$\text{term}(j=2) = -0.19 \times 0.0360 \times 0.80 \times 0.004455 = -2.438 \times 10^{-5}$$

(6 marks: correct γ_2 and $j = 2$ term.) Total drift:

$$\text{drift of } L_0 = -2.357 \times 10^{-5} - 2.438 \times 10^{-5} = \boxed{-4.794 \times 10^{-5}}$$

(2 marks: correct sum.) Note the summation runs $j = 1, 2$ (i.e. $j = i + 1, \dots, N - 1$ with $i = 0, N = 3$), exactly the forward-in-tenor range Chapter 13 flags as the most common implementation error to get backwards.

Solution 8 (10 marks). Rebonato's form:

$$\sigma(2.5) = (0.04 + 0.25 \times 2.5) e^{-1.00 \times 2.5} + 0.08 = (0.04 + 0.625) \times e^{-2.5} + 0.08$$

$$= 0.665 \times 0.082085 + 0.08 = 0.05459 + 0.08 = 0.13459 \rightarrow \boxed{13.46\%}$$

(7 marks: correct evaluation.) The long-run floor as time-to-expiry $\rightarrow \infty$ is $d = 8\%$, read directly off the form without recomputation, since $e^{-cx} \rightarrow 0$. (3 marks: correctly stated as d , with the reasoning that the exponential term vanishes.)

Solution 9 (10 marks). Total variance across all 10 factors is the sum of all eigenvalues:

$$7.35 + 1.42 + 0.61 + 0.62 = 10.00$$

(3 marks: correct total, and recognising it should equal 10 — the number of variables — for a correlation, as opposed to covariance, matrix, which is a useful self-check.) Fraction explained by the first three:

$$\frac{7.35 + 1.42 + 0.61}{10.00} = \frac{9.38}{10.00} = 0.938 \rightarrow \boxed{93.8\%}$$

(4 marks.) Yes, a 3-factor reduction is well justified on this evidence: 93.8% is comfortably above the conventional 90–95% threshold used in practice (Chapter 14), and the sharp drop-off after the third eigenvalue (from 1.42 down to a remaining seven eigenvalues summing to only 0.62, each individually small) indicates the remaining factors are capturing noise rather than systematic curve movement, not that a fourth or fifth factor is being unjustifiably discarded. (3 marks: correct recommendation with reasoning referencing the eigenvalue drop-off, not just the percentage threshold in isolation.)

